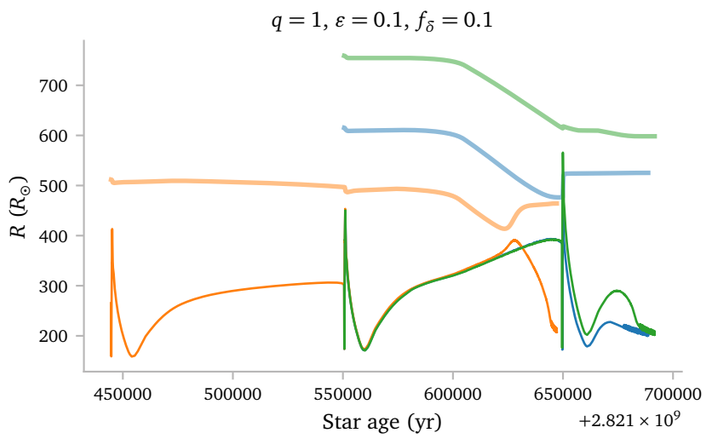
Koen Essers • Master Internship • October 17, 2026

Notes Week 33

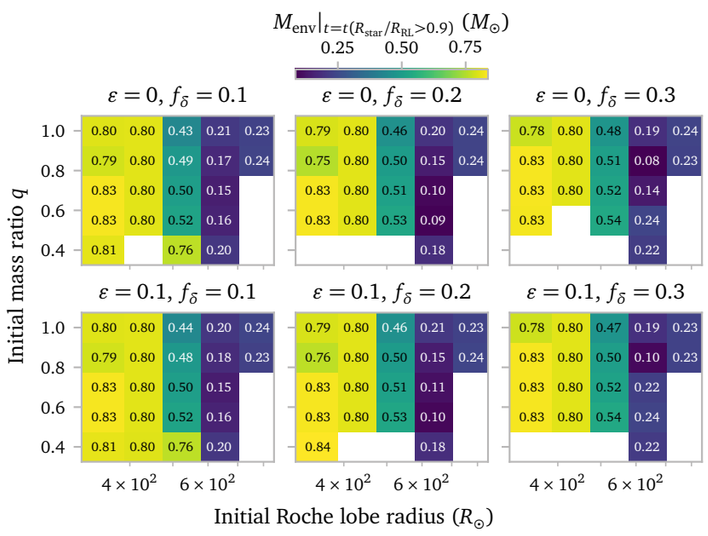
a. Envelope mass

Last week, I showed a couple of plots in which RLOF occured during the interpulse period of the TPAGB star. When the mass ratios were not extreme, and not a lot of mass was lost to a circumbinary disk, RLOF appeared to be very stable[1]. For example, see this figure:



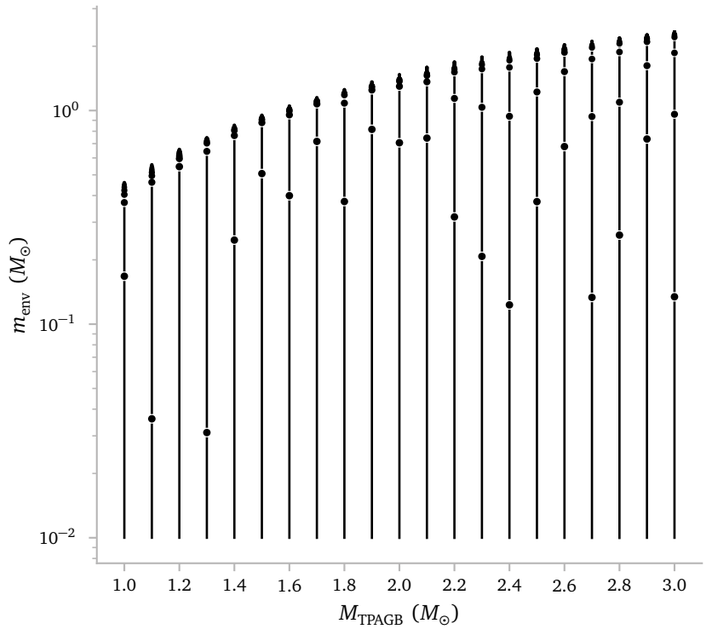
1 In the sense that the radius of the star never became larger than the Roche lobe radius.

Maybe this surprising stability is due to the low envelope mass at the moment of RLOF. Below, I plot an approximate mass for the start of RLOF. It is approximate because it is difficult to select an exact starting moment. This is also why there are differences in the last two columns.



The third column shows models that interact during the last interpulse phase, where the radius of the star never exceeds the Roche lobe radius. As we can see, the envelope mass at this point is already quite small. But really, to know if this is really the reason, we should compare if with stars with different masses, but equally small evelope masses before the final thermal pulse.

Let me plot the masses of the envelope for the different thermal pulses as a function of the mass.



The dots represent the mass of the TPAGB star at the start of a certain thermal pulse. The exact thermal pulse numbers are irrelevant here. It is only important to look at the lowest two dot. These dot represent the mass of the envelope during the *last* two thermal pulse. Clearly, if we interact during the last *interpulse* phase, RLOF will occur when the star has an envelope mass between these two values.

The figure above shows that for the models that interact during the interpulse, the envelope mass is approximately 0.5 $M_\odot$. This figure shows that this is totally in agreement with the values that we expect are possible. The envelope mass of the 1.4 $M_\odot$ star during the last thermal pulse is not very low compared to the other stars.

There are some quite heavy stars that have *lower* envelope masses towards the end of the last interpulse period. For example, it might be interesting to look at the 2.4 $M_\odot$ model. This one has an envelope mass during the last thermal pulse close to 0.1 $M_\odot$, about 2× lower than the 1.4 $M_\odot$ model.

I will use the simple binary code to estimate a nice Roche lobe radius that hits causes RLOF to happen towards the end of the last interpulse, when there is very little mass left. It is then interesting to see if we again obtain these stable RLOF mass transfer binary systems.

References